

DNA Damage and Repair

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Objectives

- Identify different causes of DNA damage.
- Identify sources of DNA damage.
- Mention 4 different mechanisms of DNA repair.
- Recognize three different inherited disorders of defective DNA repair

Introduction

- DNA is constantly being altered by various endogenous or exogenous processes, including cellular division
 - DNA replication is accurate process
 - Causes an error 1×10^{-7}
- Maintenance of the genetic information requires constant repair
- All living organisms have mechanisms to repair damage to DNA.

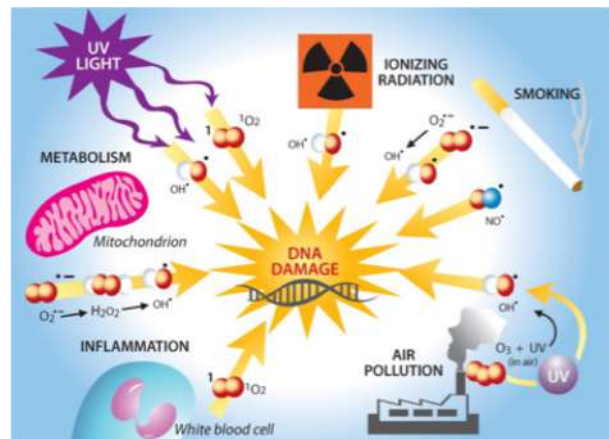
DNA Damage

- DNA Damage
 - Is an abnormal change in the chemical structure of the DNA.
 - Occurs outside the timeframe of replication.
 - In the context of this lecture, it also includes DNA mutation
 - a change in the sequence of base pairs
- 10,000 -1,000,000 nucleotide/cell/day

Sources of DNA damage: Endogenous sources:

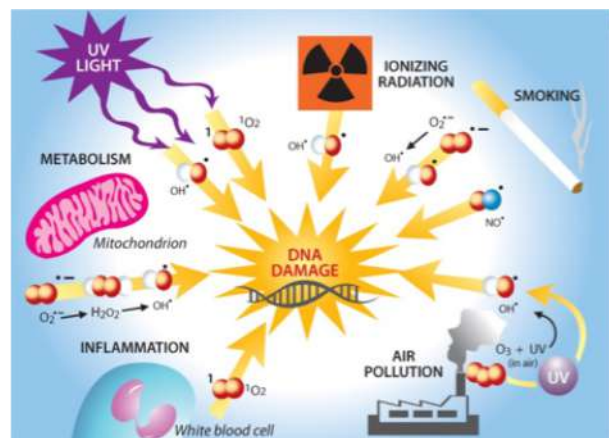
Arises from normal cellular metabolism:

- Oxidation: Reactive Oxygen Species (ROS)
- Inflammation
- Hydrolytic reactions:
 - Depurination: breakage of the glycosyl bond between purines (A/G)
 - Deamination: C – U
- DNA methylation (SAM)
- Replication and mitotic stress.



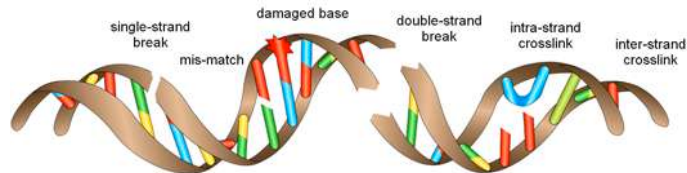
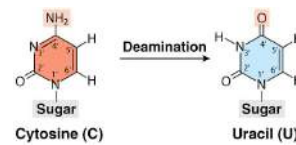
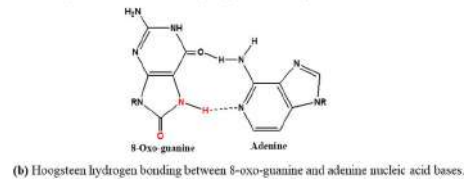
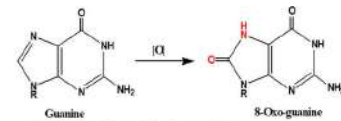
Sources of DNA damage: Exogenous sources (Mutagens):

- Radiation (ionizing or ultraviolet)
- Various chemicals, heavy metals, chemotherapeutic drugs
- Air pollutants, and inflammatory responses.
- Alkylating agents (tobacco smoke) [?] addition of ethyl or methyl to a nucleotide (Guanine)



Examples of DNA damage

- Oxidation of Bases
 - Reactive oxygen species
 - -OH radical the most reactive and capable of damaging DNA, protein and lipids
 - Leads to G to T conversion
- Spontaneous Base Deamination
 - Leads to C to U conversion
- Single/Double strand breaks
 - UV, Ionizing radiation, ROS, **stalled replication fork**

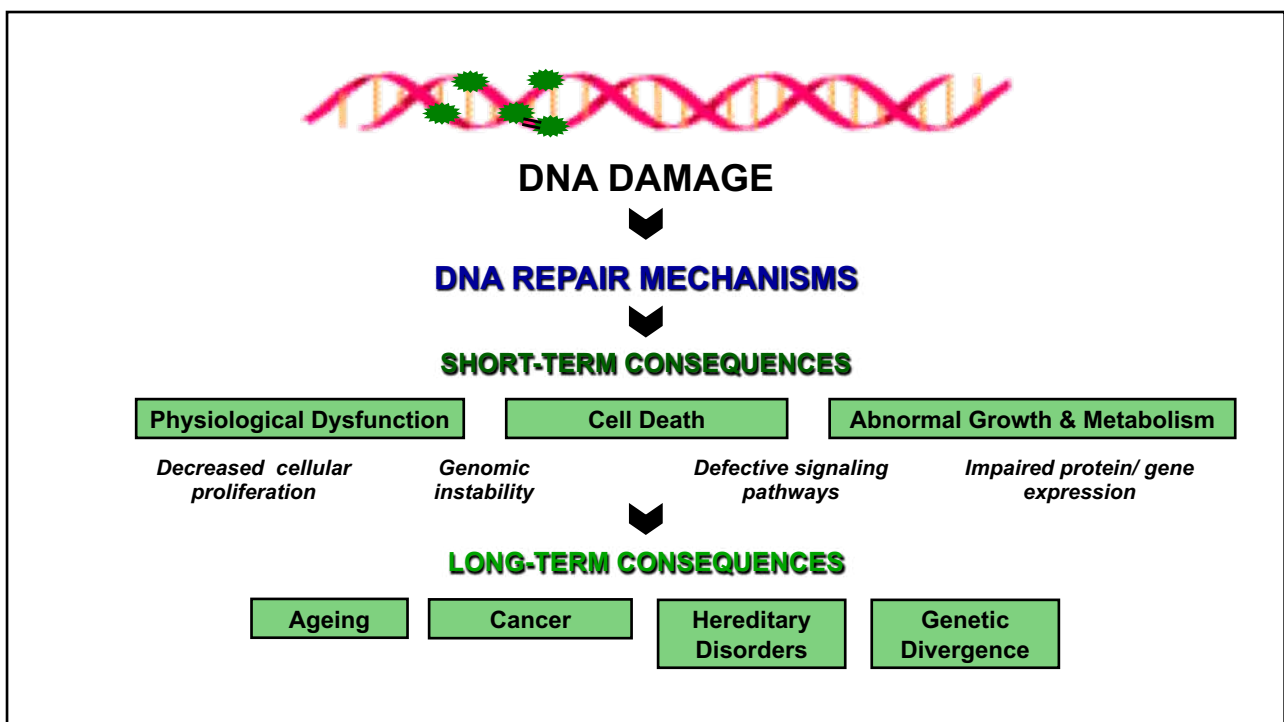


DNA repair

- DNA repair
 - is a collection of processes by which a cell identifies and corrects damage to the DNA molecules that encode its genome.
 - Cells signaling pathways allow cell-cycle arrest to provide increased time for DNA repair.
- Most errors are removed and corrected by DNA repair enzymes
 - Process of DNA proofreading
 - Corrects 99.9% of errors
- If the damage is irreparable → cells may initiate programmed cell death (apoptosis).
 - Mutations arise from the cell's attempt to tolerate damage

Consequences of failed DNA repair

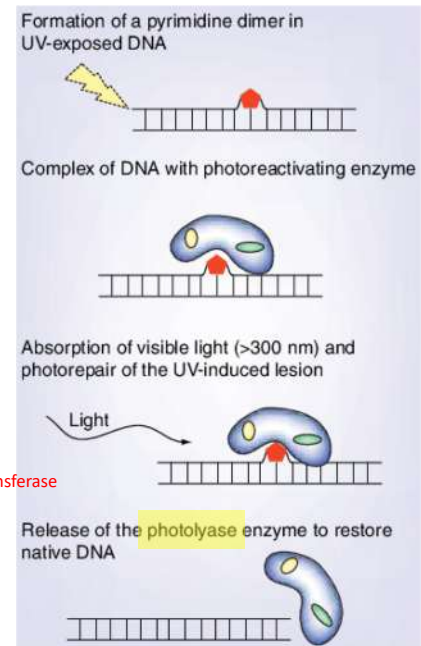
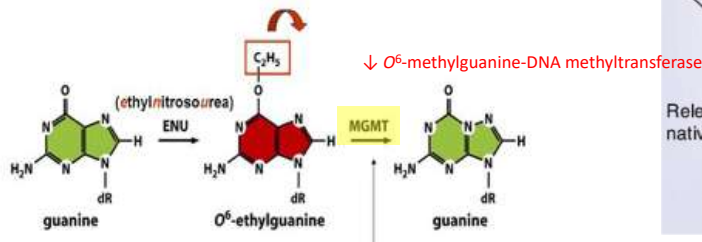
- Genomic instability, apoptosis, or senescence
 - Can greatly affect the individual's development.
- Predisposes to:
 - Immunodeficiency
 - Neurological disorders
 - Cancer



DNA Repair Mechanisms

• Direct Reversal

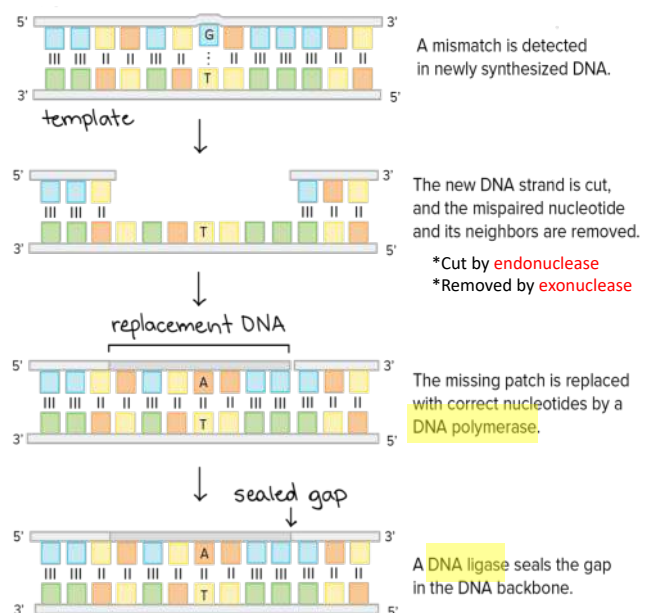
- Single step process, does not require a template.
 - Demethylation → Methyl Group Removal
 - DNA methylation of imprinted loci observed in many cancer types
 - Photoreactivation → in bacteria
 - Repairs pyrimidine dimers



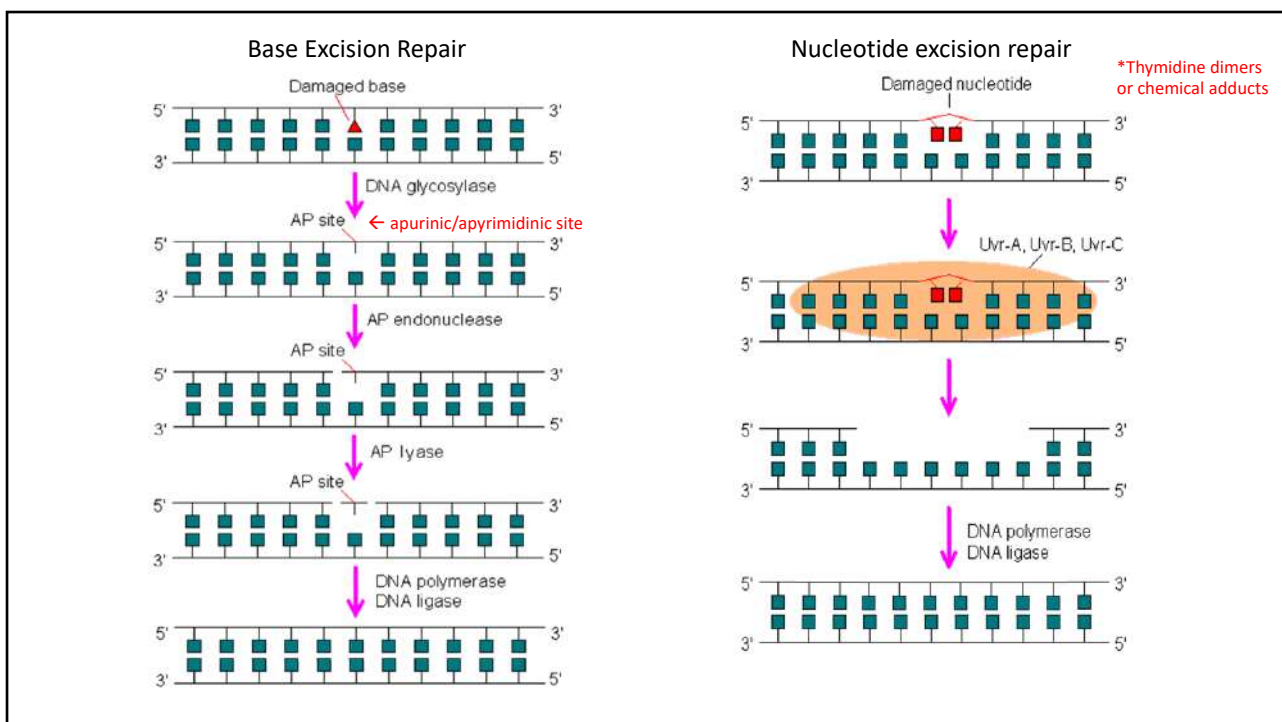
DNA Repair Mechanisms

• Single strand repair mechanisms

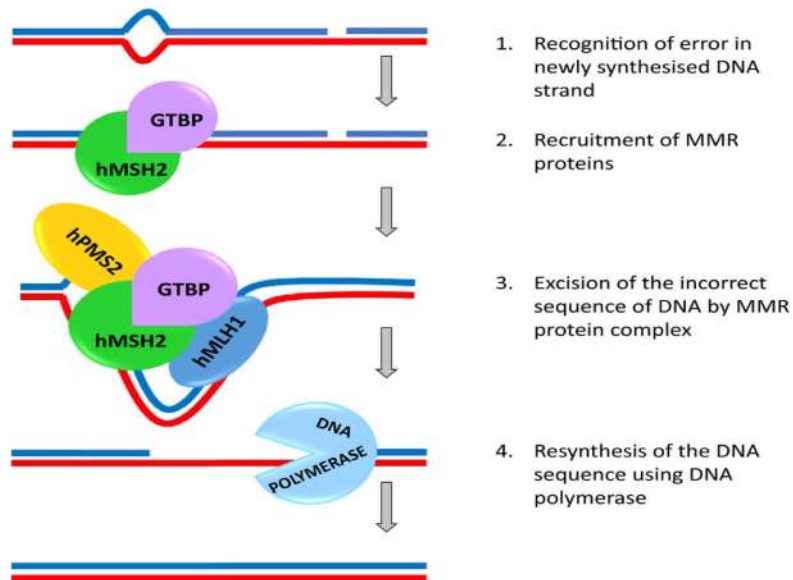
- Multistep processes
- Utilizes undamaged strand as a template
 - Excision repair
 - Base excision repair (BER) mechanism
 - Nucleotide excision repair (NER) mechanism
 - Mismatch repair (MMR) mechanism



- **Base excision repair (BER)** corrects those forms of oxidative, deamination, alkylation and abasic single base damage that are not perceived as significant distortions to the DNA helix.
- **Nucleotide excision repair (NER)** is the pathway of choice to remove bulky lesions such as thymine dimers and lesions from UV radiation, benzo[a]pyrene adducts, or damage from chemotherapeutic agents.
- **Mismatch repair (MMR)** is a post replicative repair pathway. Typical substrates for the MMR pathway are base mismatches that have arisen during replication and the insertion-deletion loops (IDLs) within repetitive DNA sequences that have resulted from strand slippage events.

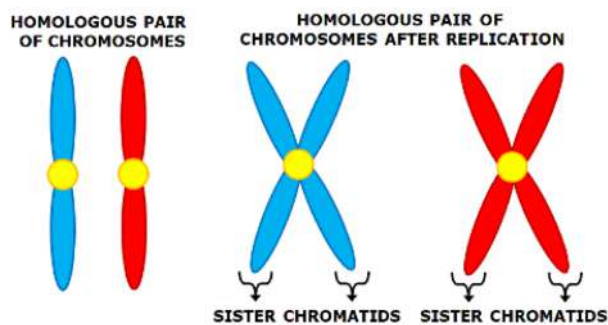


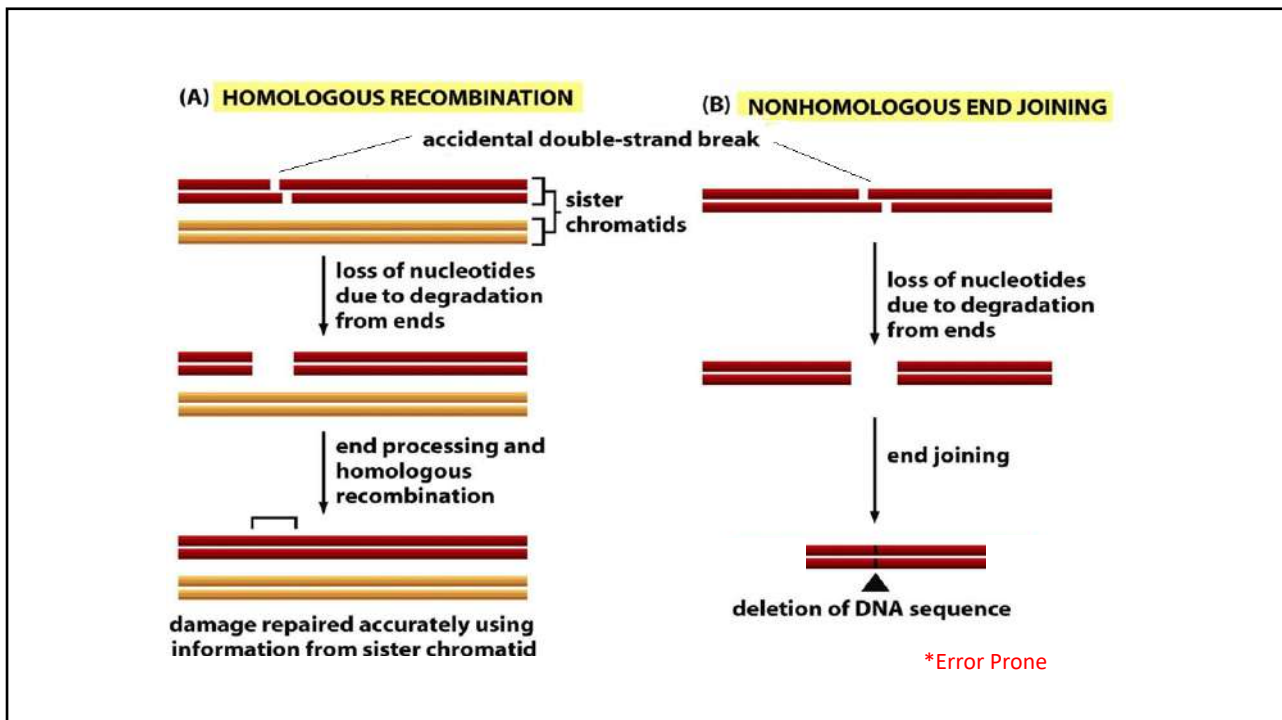
Mismatch repair (MMR) mechanism



DNA Repair Mechanisms

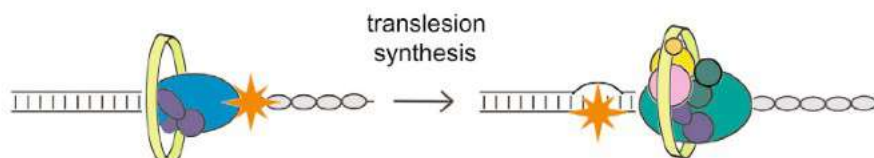
- Double strand repair mechanisms
 - Homologous recombination (HR) mechanism
 - Non-homologous end joining (NHEJ) mechanism

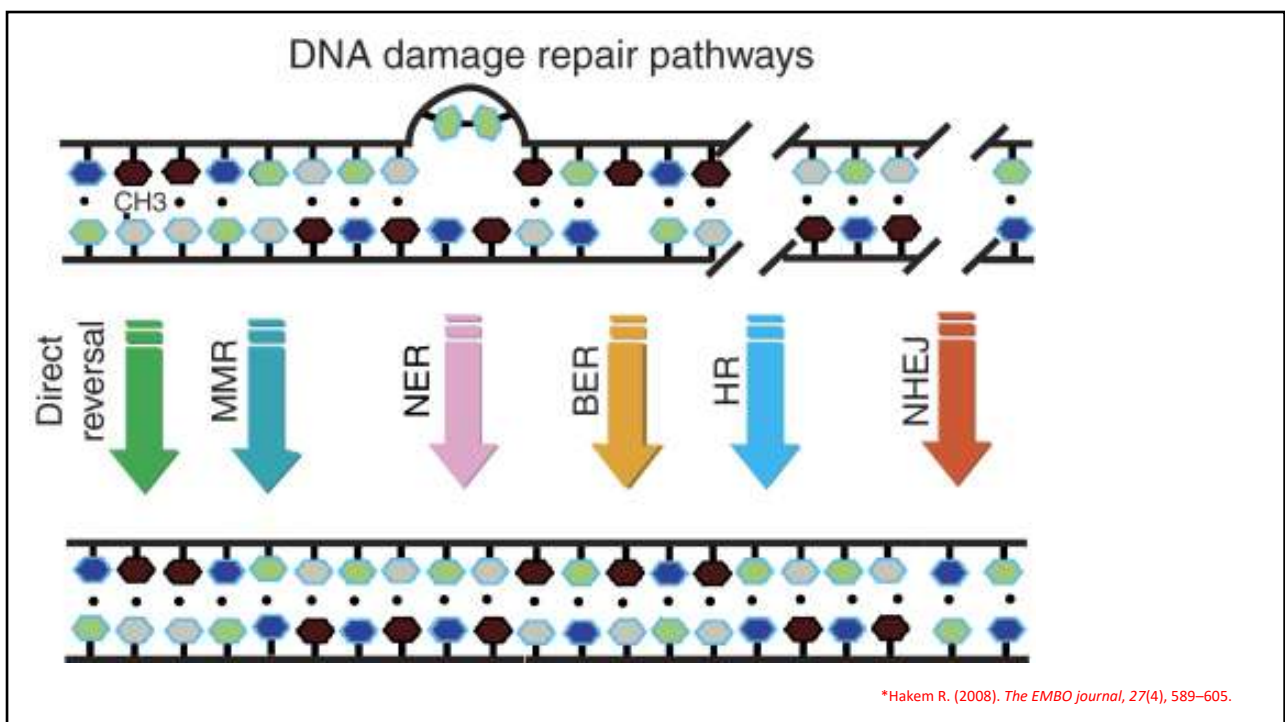
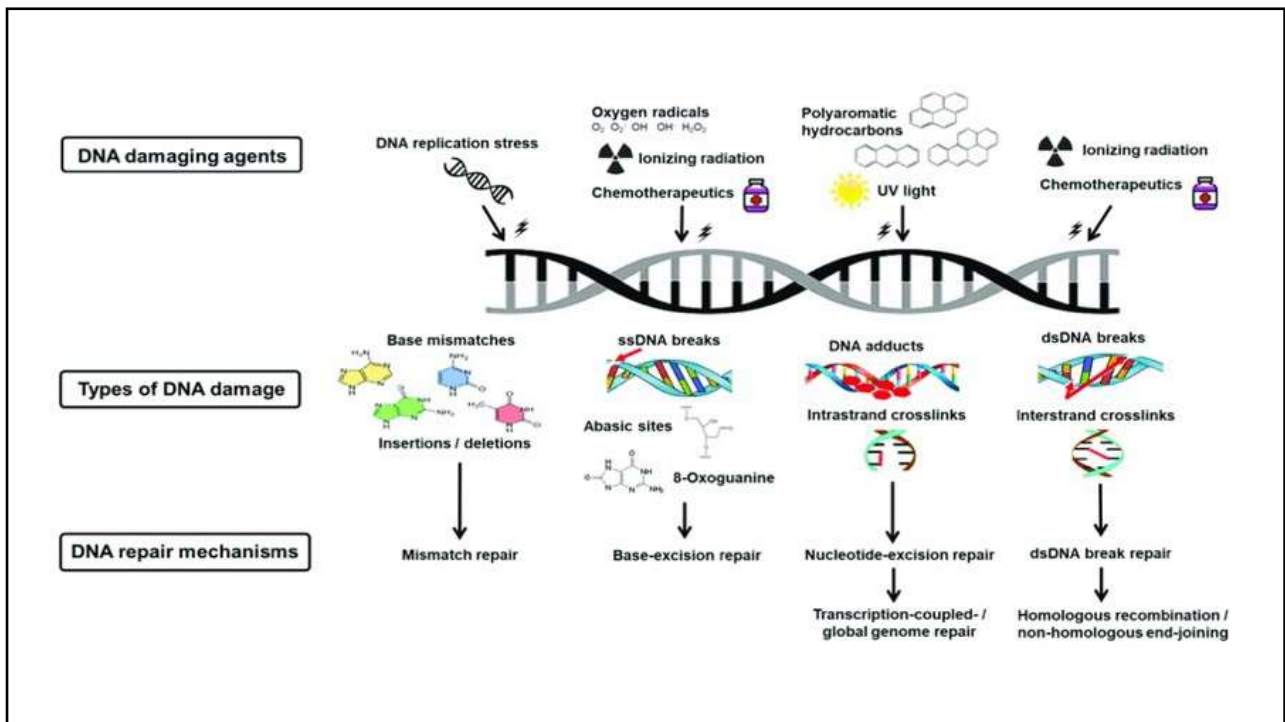




Trans-lesion DNA synthesis

- Involves specialized DNA polymerases
 - Able to bypass DNA lesions and synthesize DNA across the damage site
 - Allowing replication to continue
 - Why? If replication stalled, may lead to double strand breaks
- if incorrect nucleotides were incorporated by TLS polymerases, they would become mutations in the next round of replication, which propel tumorigenesis and disease, but can also contribute to the overall fitness and evolution of organisms.





*Hakem R. (2008). *The EMBO journal*, 27(4), 589–605.

Disorders of DNA repair

Table 2.7

DNA repair pathways, genes, and associated disorders

Type of DNA Repair	Mechanism	Genes	Disorders
Base excision repair (BER)	Removal of abnormal bases	<i>MUTYH</i>	Colorectal cancer
Nucleotide excision repair (NER)	Removal of thymine dimers and large chemical adducts	<i>ERCC</i>	Xeroderma <i>COFS</i> pigmentosum
Postreplication repair	Removal of double-strand breaks by homologous recombination or non-homologous end-joining	<i>NBS</i>	Nijmegen breakage syndrome
		<i>BLM</i>	Bloom syndrome
		<i>BRCA1/2</i>	Breast cancer/ <i>Fanconi Anemia</i>
Mismatch repair (MMR)	Corrects mismatched bases caused by mistakes in DNA replication	<i>MSH</i> and <i>MLH</i>	Colorectal cancer (HNPCC)

HNPCC , Hereditary non-polyposis colorectal cancer.

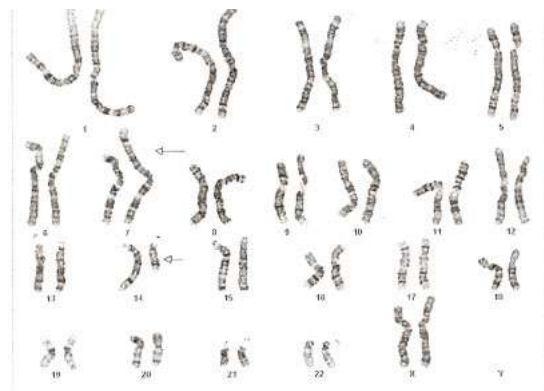
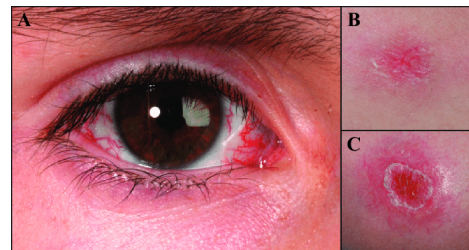
TABLE 42-1 Representative DNA Repair Disorders

Pathway	Representative Condition	Major Clinical Features
Nucleotide excision repair	Xeroderma pigmentosum	Extreme sunburns, early onset of skin cancers and photo-aging of skin
Base excision repair	MUTYH-associated polyposis	Development of hundreds of adenomatous polyps and early onset colon cancer
Mismatch repair (See Table 42-2 for description of biallelic form)	Lynch syndrome (hereditary non-polyposis colon cancer)	Increased risk and earlier onset of colon, endometrial, ovarian, and biliary tract cancers
Double strand break recognition and repair	Ataxia-telangiectasia	Cerebella ataxia, telangiectasia, immunodeficiency, leukemias, lymphomas, and solid tumors
Cross-link repair and homologous recombination	Fanconi anemia	Bone marrow failure, congenital anomalies, abnormal pigmentation, and increased risk of leukemias and head and neck and genitourinary malignancies
RECQ helicase disorders <i>*Unwinds double strands</i>	Bloom syndrome	Short stature, butterfly rash on face, immunodeficiency, and increased risk of multiple cancer types

Ataxia Telangiectasia

- Clinical Features
 - Conjunctival telangiectasia
 - Ophthalmoplegia
 - Immunodeficiency
 - Ataxia/chorea
 - Radiation sensitivity
 - Leukemia / lymphoma
- Diagnosis: *ATM* gene mutation
 - High frequency of 7;14 translocation or chr. 7 /14 pericentric inversion in t-lymphocytes

TP53 "Guardian of the Genome" and ATM
 → Required for sensing of DNA damage
 and regulating DS DNA repair proteins



Xeroderma Pigmentosum

- Extreme sensitivity to light
 - Freckling, pigmentary changes, cancers (BCC, SCC, melanoma), skin thinning and dryness
- Neurological
 - Deafness, ataxia, spasticity and intellectual disability
- Corneal clouding and other ocular tumors



Defect in nucleotide excision repair
 -Extreme sensitivity to UV light & a high frequency of skin cancer

Fanconi Anemia

- Common features
 - Café au lait macules, short stature, Radial ray anomalies
 - Pancytopenia, aplastic anemia
 - Renal malformations
 - Cancers
 - Myelodysplasia
 - Squamous Cell Carcinoma
 - Female reproductive tract cancers
- Diagnosis
 - chromosomal breakage



Interstrand cross-link repair defect →

These lesions are recognized and repaired by the Fanconi anemia (FA) proteins



chromosome breakage assay is widely used for the primary diagnosis of FA.

THANK YOU
Questions?

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